

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION  
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE  
OR BRAND NAME

**PICK ONE**

**Sole Source:**

☒ I recommend that NASA, NASA Shared Services Center negotiate with Vedas Tech LLC (Vedas Engineering) only for Specialized Multiphysics Training. The total estimated cost of this effort is ██████████ and the estimated period of performance or lead-time for delivery is April 23, 2025 – June 13, 2025.

**OR**

**Brand Name Justification:**

☐ I recommend that NASA, NASA Shared Services Center negotiate with only with vendors who can provide the subject band name from Manufacture Name for Enter Supply or Service to be procured. The total estimated cost of this effort is Enter Dollar amount and the estimated period of performance or lead-time for delivery is Date.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacture. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

MARM+ couples heat transfer, thermal expansion, structural stress, fluid mechanics, hydrogen migration (only for hydrogenated moderators) and nucleonics, and has the potential to incorporate transient system modeling. At present, we have no such capability that is mature. This will help us to potentially “jump the queue” in terms of our insight/in-line work. The training course is concentrated into three sessions over ten days total, allowing for relatively rapid assimilation and evaluation by our SMEs. The training is provided by the lead developer and so is likely the best training available. The training will be more efficient than the current approach which does not provide access to the software developer and more advanced methods. Regardless of our future with MARM+, this will likely provide us with M&S techniques for coupling in-house tools, as it already has by providing us a method for incorporating unstructured meshes into MCNP through informal engagement with the software.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

The impact to the Government includes schedule impacts and technical capability impacts. Regarding schedule, there is no known viable alternative that can be deployed immediately. This method (a) uses tools already known to the team, (b) is mature, and (c) provides direct access to the most-knowledge developer and user of the software. Pursuing another method would require in-house development, pursuit of a method that does not accord with known tools or that requires significant development to be brought to use to fulfill modeling and simulation (M&S) goals of insight and in-line work, adding six to eight months of delay to identify an equivalent program (which is not known to exist) with indeterminate results. With the current M&S plan and associated significant milestones, it is not possible meet our schedule with an alternative vendor. Technically, the ability of NASA to adequately perform insight and in-line work in the most critical area of nuclear core multiphysics and engine system modeling for a high-visibility project – SENTRE (formerly DRACO) will remain inadequate to answer important design

questions, such as the effects of thermal expansion on nuclear power feedback. Speed is essential for this particular project given that the intended target for deployment of this capability (SENTRE) is transitioning to a NASA-led project presently, requiring multiphysics capabilities to be fully capable as soon as possible (current target is end of May 2025) so that NASA can be fully-informed technically about the behavior of the heart of the system – the nuclear core.

**ADDITIONAL COMMENTS:**

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[Redacted]

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**Date**

[Redacted]

**Title**

[Redacted]

**Office**